How rich is the EEG signal?

A hands-on introduction to time-resolved decoding

A bit about my (brief) academic journey

- BSc in Psychology (Developmental track) @ UniPd
- MSc in Psychology (Neuroscience track) @ UniTn
  - MSc thesis @ MPI CBS
- Currently, in my 3rd year of PhD in Linguistics @ IUSS Pavia
  - Visiting research period @ Goethe Universität

A bit about our time together



Theoretical introduction

- The EEG signal
- The signal-to-noise ratio
- Approaches to EEG analysis
- Predictive modelling framework
- Introduction to the dataset

A bit about our time together

10^{00} — 10^{45} — 11^{30}

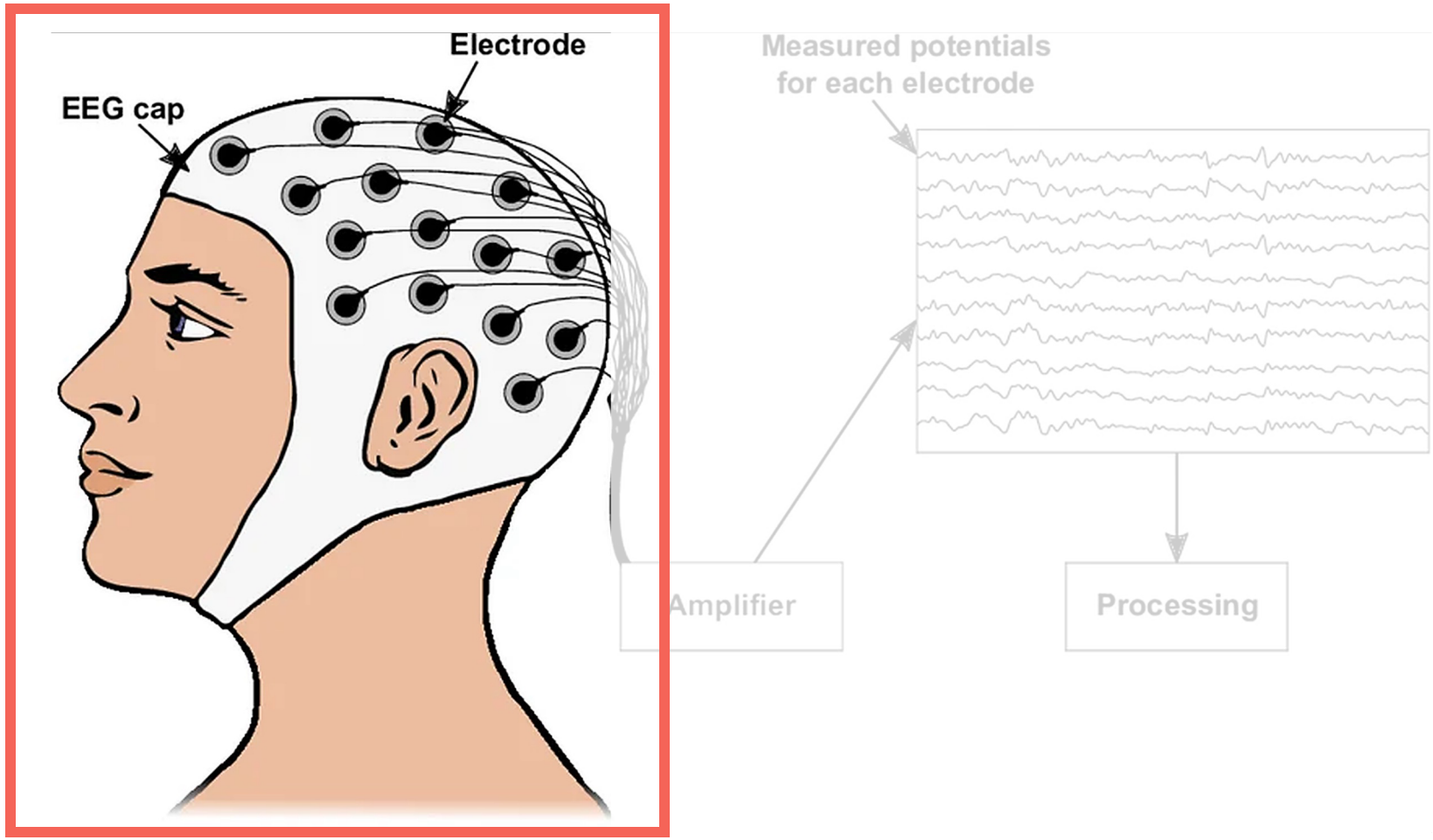
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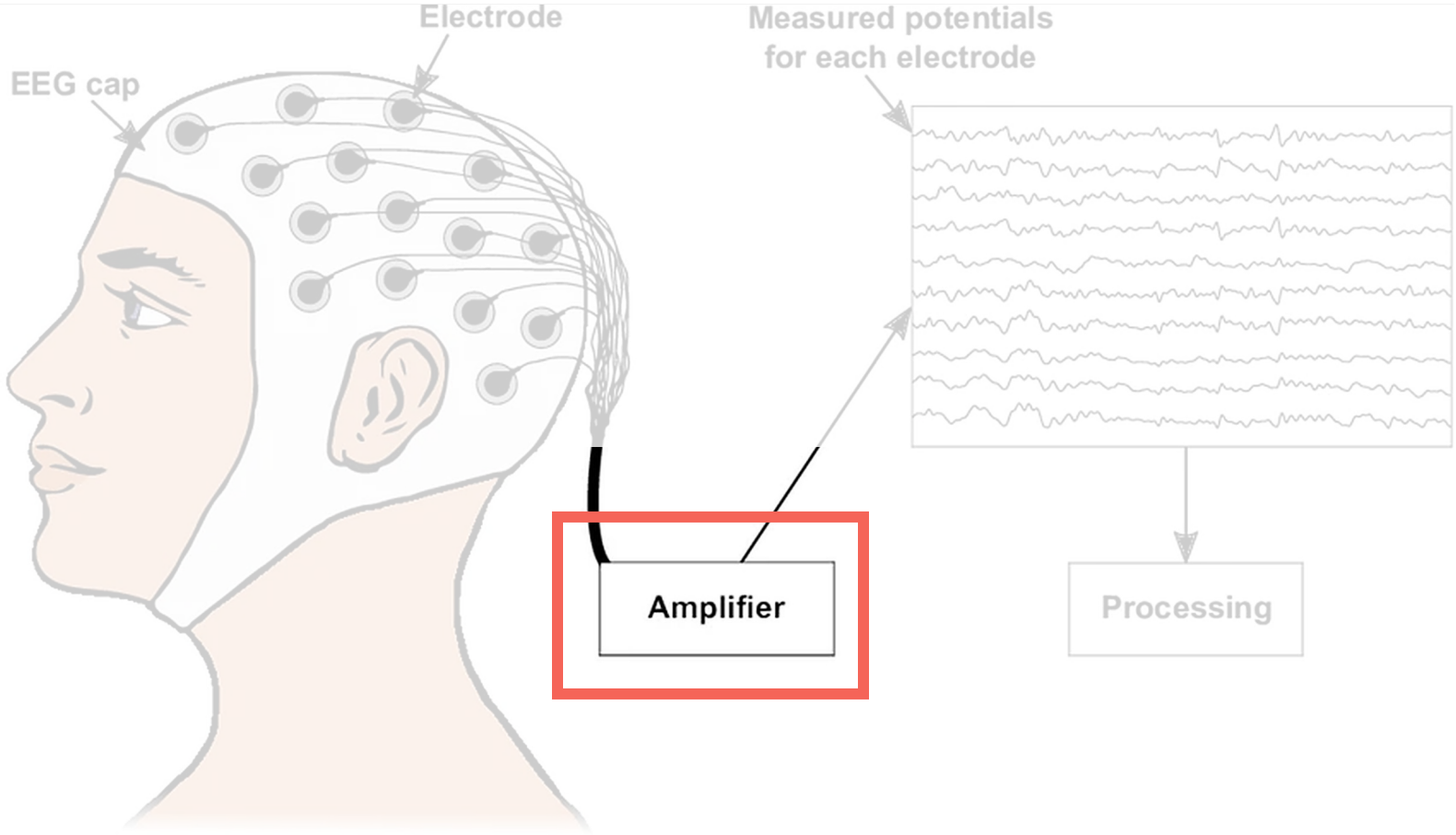
Hands-on session

- Colab environment
- Data inspection
- Time-resolved regression

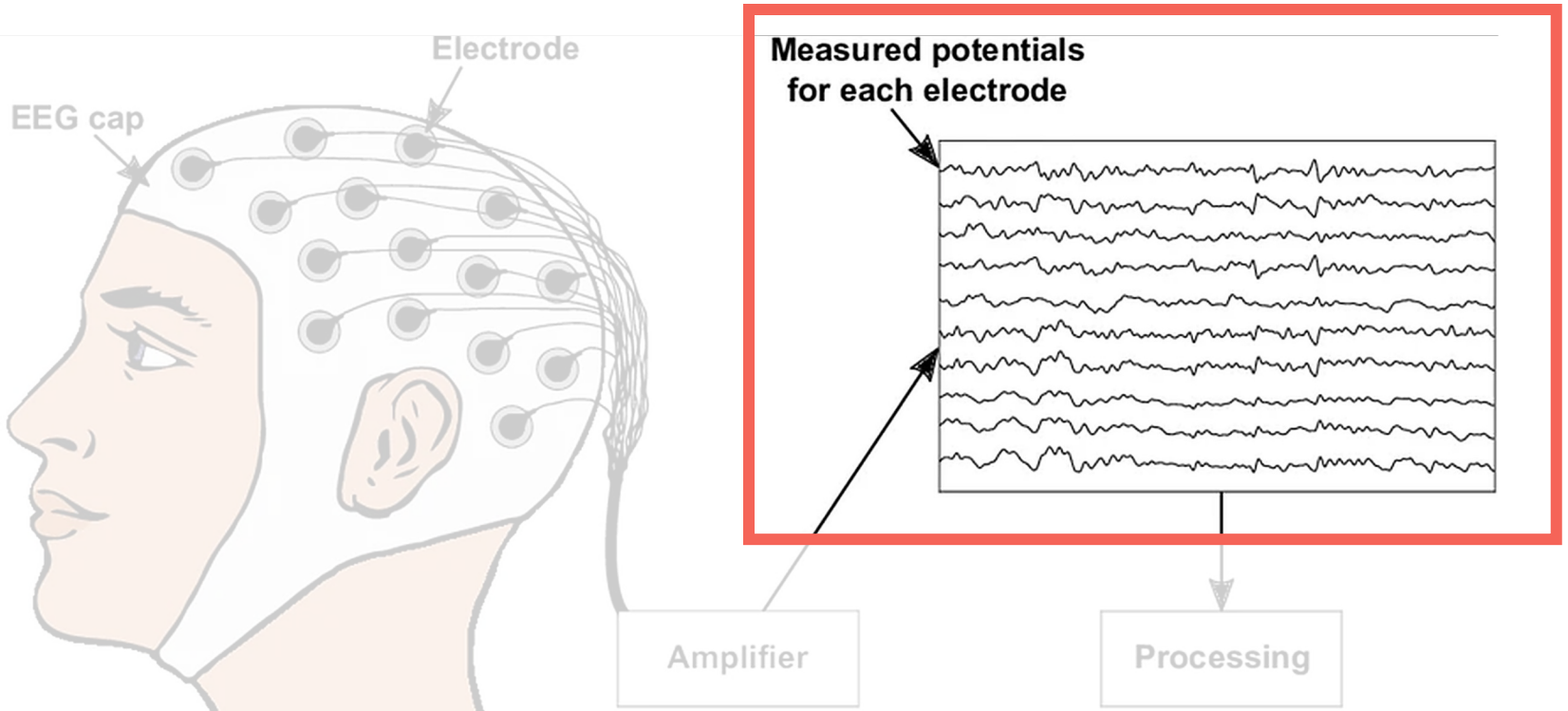
The EEG signal



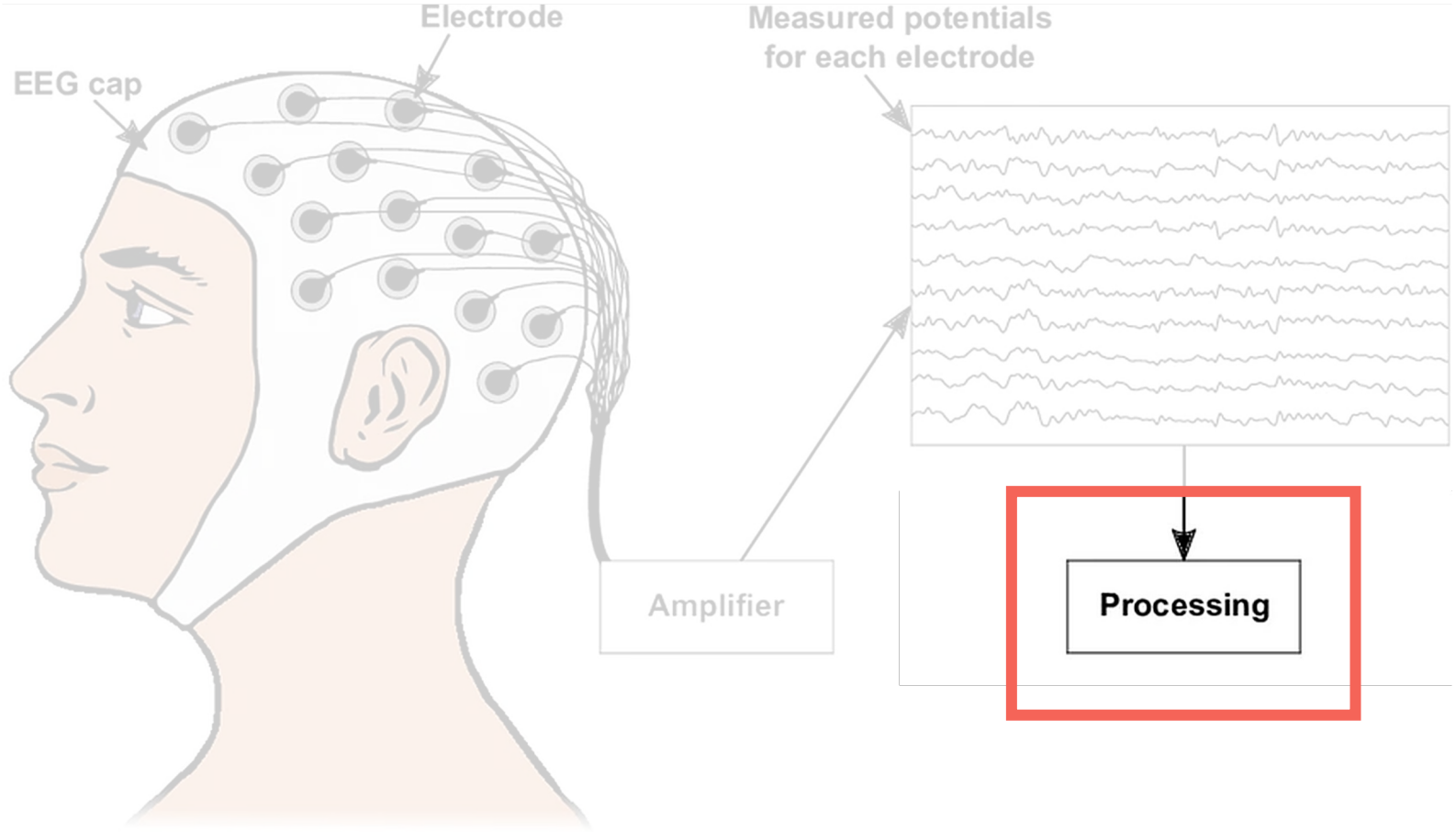
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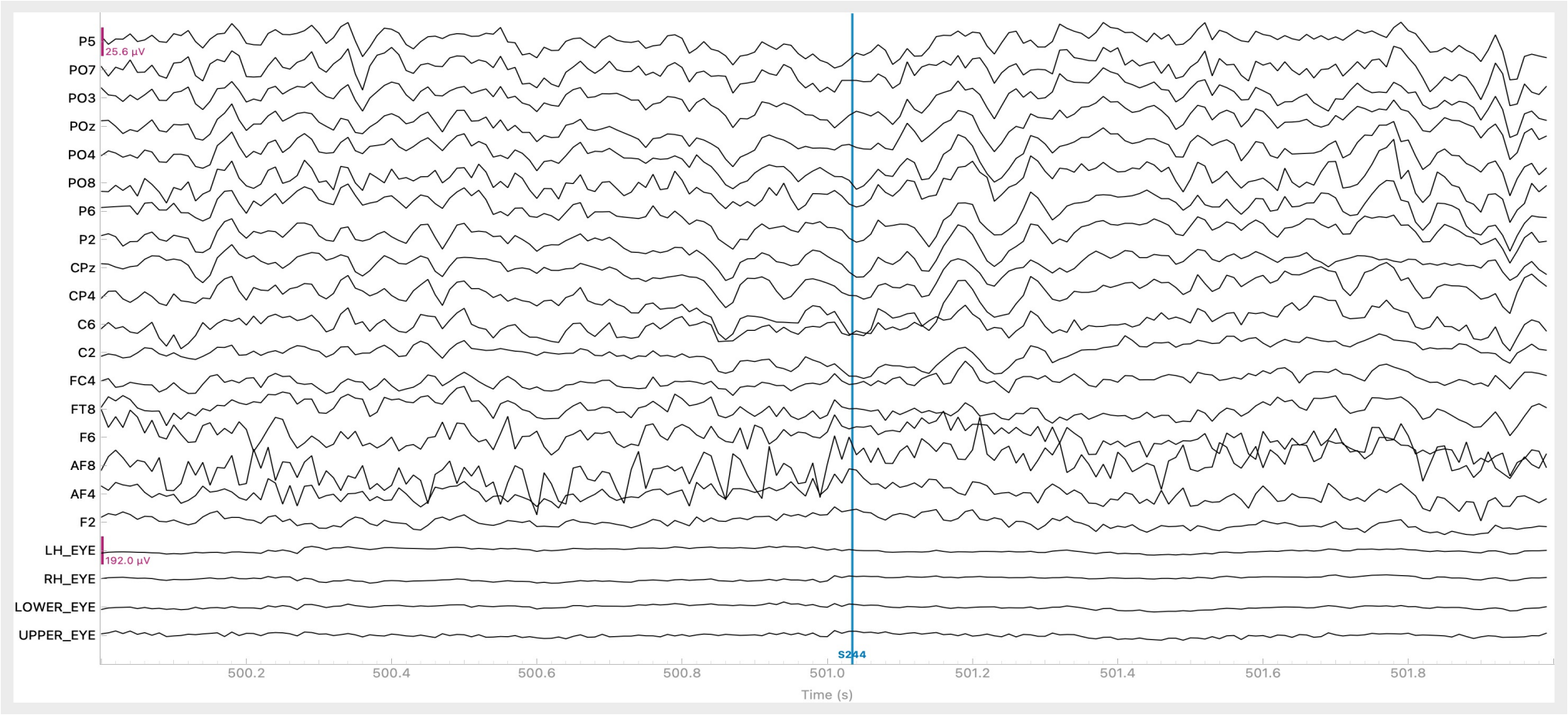
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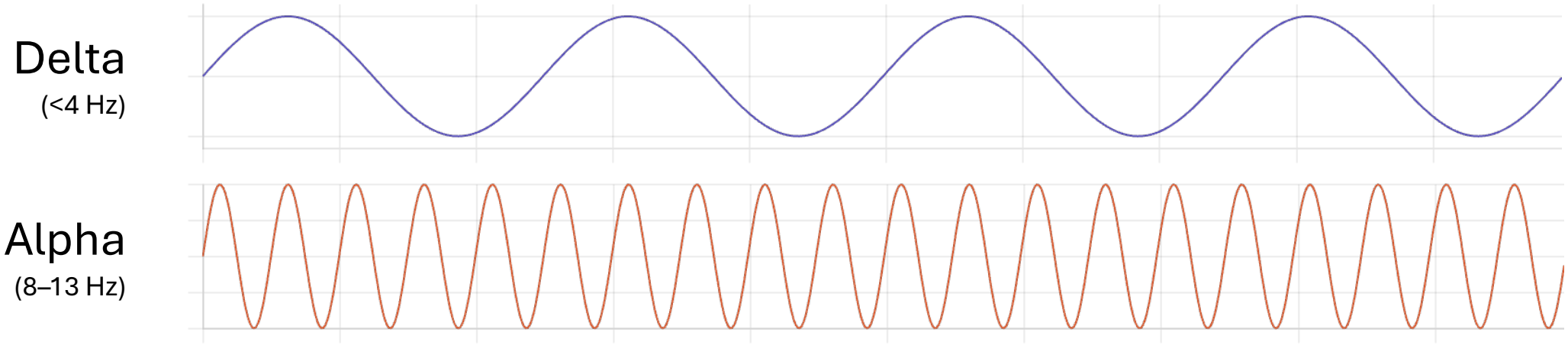
The EEG signal



The EEG signal

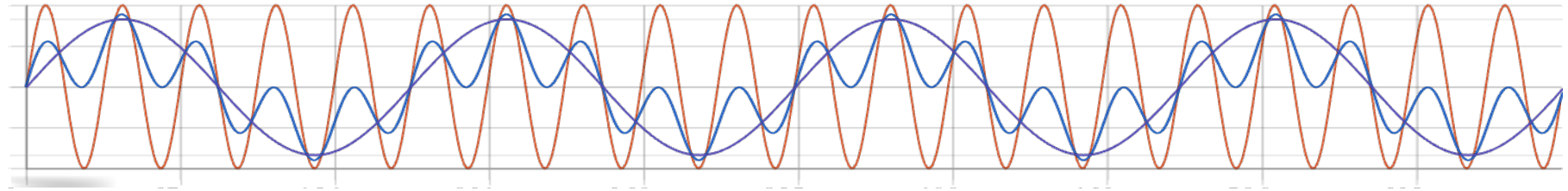


The EEG signal

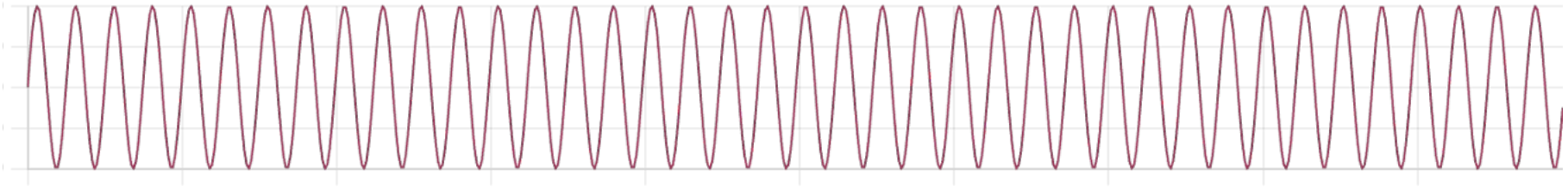


The EEG signal

EEG



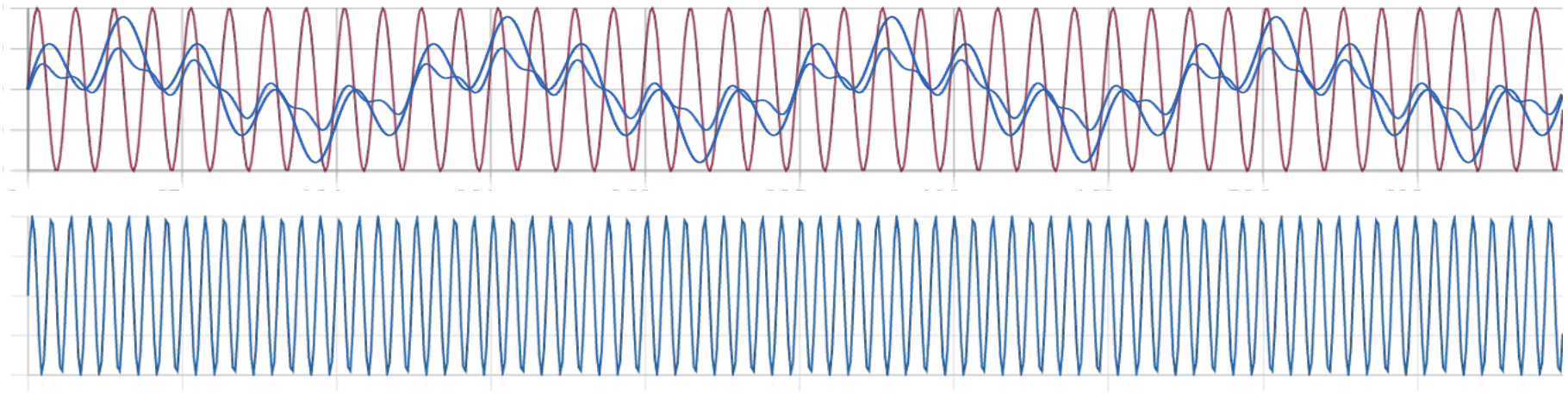
Beta
(13–30 Hz)



The EEG signal

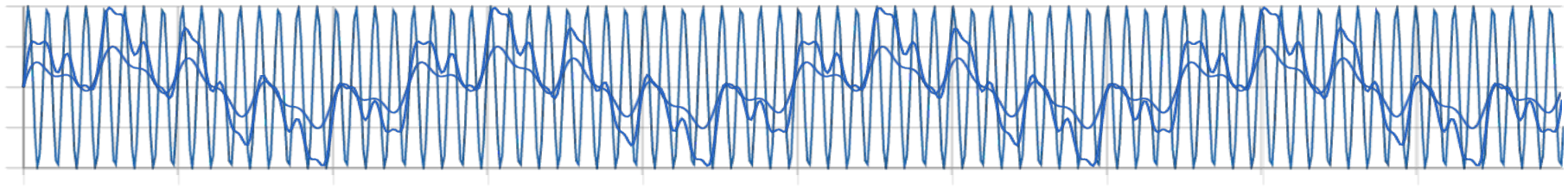
EEG

Gamma
(>30 Hz)



The EEG signal

EEG



The EEG signal

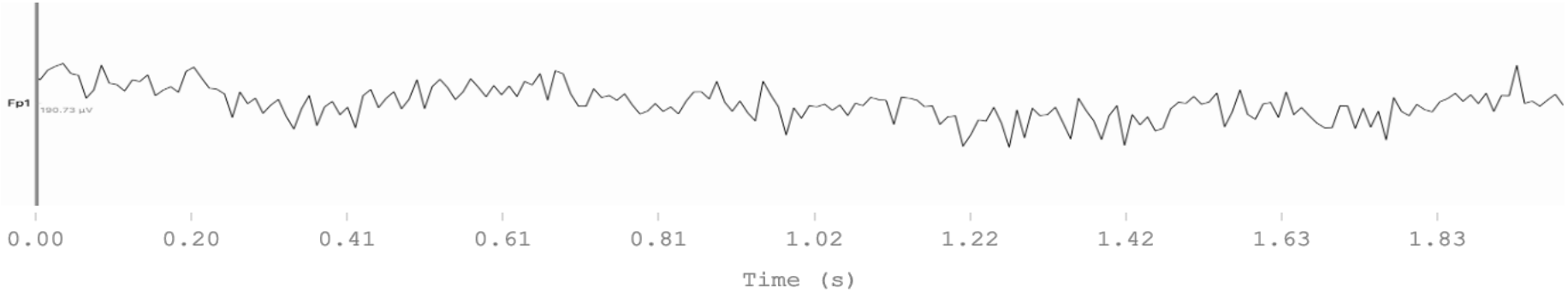
- Synthetic data

EEG



- Real data

EEG



The signal-to-noise ratio

- Signal-to-noise ratio, also referred to as SNR

$$SNR = \frac{signal}{noise}$$

The signal-to-noise ratio

- Signal-to-noise ratio, also referred to as SNR

neural activity of interest

$$SNR = \frac{\cancel{signal}}{noise}$$

The signal-to-noise ratio

- Signal-to-noise ratio, also referred to as SNR

neural activity of interest

$$SNR = \frac{\cancel{signal}}{\cancel{noise}}$$

unwanted background interference

The signal-to-noise ratio

- Preprocessing
    1. Filtering

The signal-to-noise ratio

- Preprocessing
    1. Filtering
    2. Re-referencing

The signal-to-noise ratio

- Preprocessing
    1. Filtering
    2. Re-referencing
    3. Bad channel detection and interpolation

The signal-to-noise ratio

- Preprocessing
    1. Filtering
    2. Re-referencing
    3. Bad channel detection and interpolation
    4. Artifact detection and correction

The signal-to-noise ratio

- Preprocessing
    1. Filtering
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    4. Artifact detection and correction
    5. Epoching

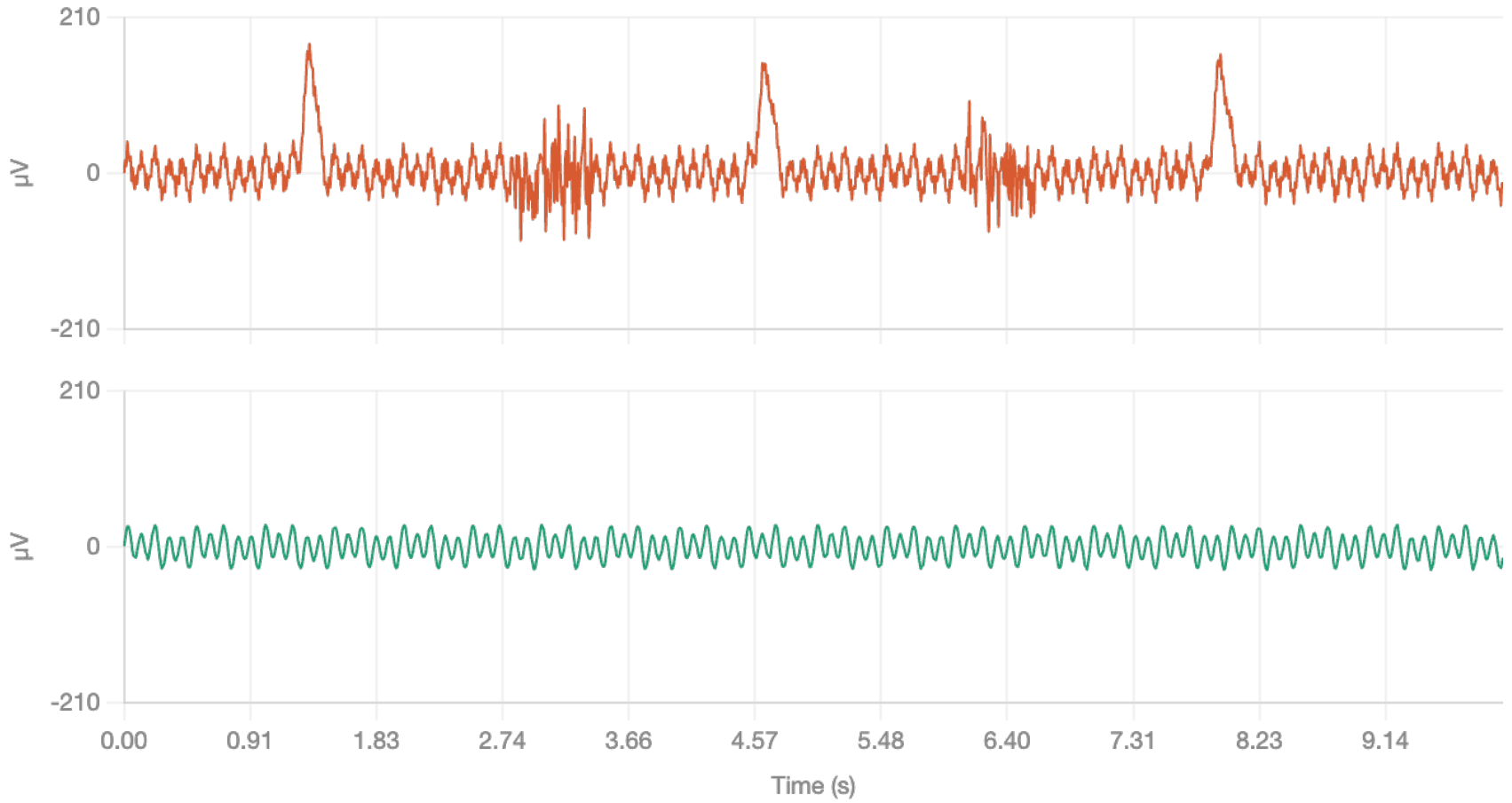
The signal-to-noise ratio

- Preprocessing
    1. Filtering
    2. Re-referencing
    3. Bad channel detection and interpolation
    4. Artifact detection and correction
    5. Epoching
    6. Averaging

The signal-to-noise ratio

(synthetic)
Raw EEG

(synthetic)
Preprocessed EEG



The signal-to-noise ratio

Want to play with basic EEG signal generation and EEG preprocessing? Do scan the QR code or visit this link.



The signal-to-noise ratio

Want to play with basic EEG signal generation and EEG preprocessing? Do scan the QR code

You have 5 minutes, starting from now

Approaches to EEG analysis

- Event-related potentials (ERPs)
- Spectral analysis
- Time-frequency analysis
- Functional connectivity
- Multivariate pattern analysis (MVPA)

Approaches to EEG analysis

Univariate

- ERPs
- Spectral analysis
- Time-frequency analysis

Bivariate

- Functional connectivity

Multivariate

- MVPA
- RSA
- PCA
- LDA

Approaches to EEG analysis

Univariate

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- Functional connectivity

Multivariate

- MVPA
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- LDA

Approaches to EEG analysis

How and why did **multivariate approaches** came about?

1. From the neuron to the population doctrine

Yuste (2015), *Nature reviews neuroscience;* Ebitz & Hayden (2021), *Neuron*

Approaches to EEG analysis

How and why did **multivariate approaches** came about?

1. From the neuron to the population doctrine
2. From “localising” activity to understanding neural codes

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Approaches to EEG analysis

How and why did **multivariate approaches** came about?

1. From the neuron to the population doctrine
2. From “localising” activity to understanding neural codes
3. Technological and computational advances

Yuste (2015), *Nature reviews neuroscience;* Ebitz & Hayden (2021), *Neuron*

Approaches to EEG analysis

There are several approaches to of multivariate analysis, answering different types of questions:

1. predictive models (i.e., decoding, encoding, temporal generalization)
2. representational geometry models (i.e., RSA)
3. latent variable models (i.e., factor analysis, ICA)
4. dimensionality reduction models (i.e., PCA)
5. brain–behaviour association models (i.e., CCA, PLS)
6. dynamical models (i.e., HMMs, SSMs)

Approaches to EEG analysis

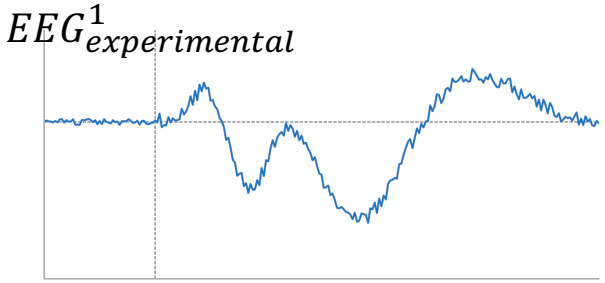
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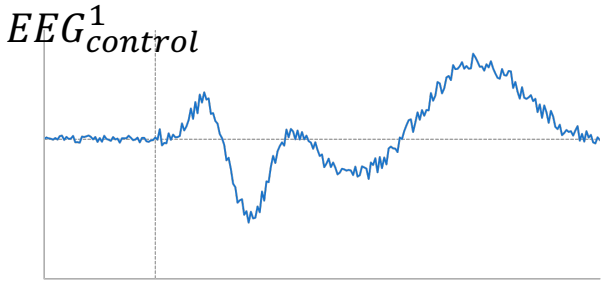
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Predictive modelling framework

Beyond the subtraction method

$$\overline{EEG}_{target} = \overline{EEG}_{experimental\ condition} - \overline{EEG}_{control\ condition}$$

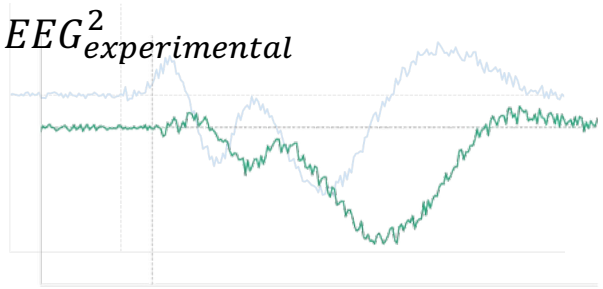


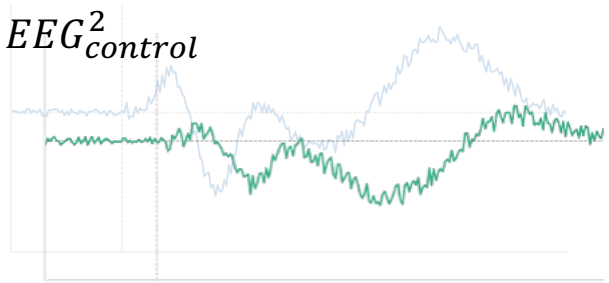


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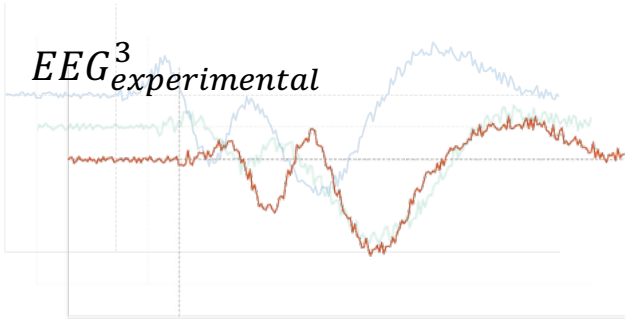


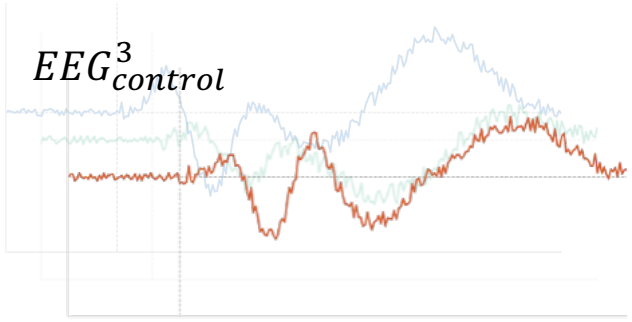


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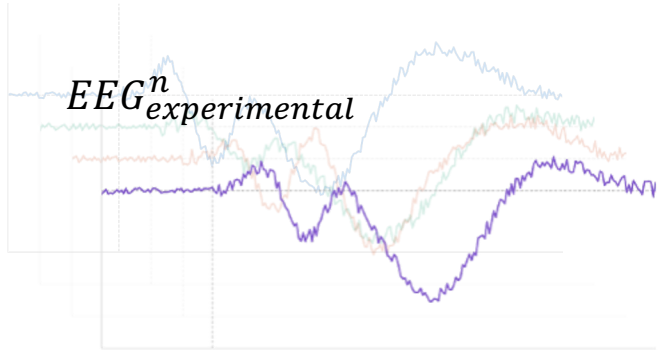


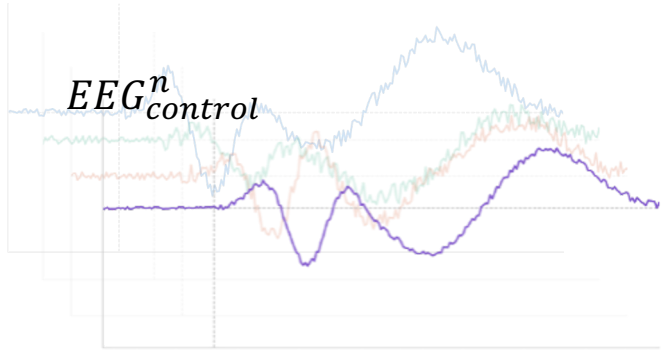


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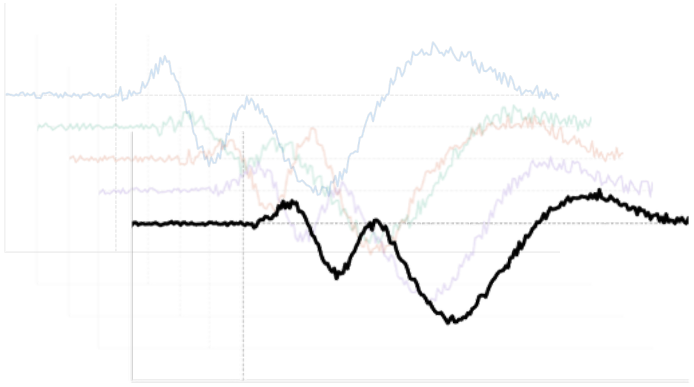


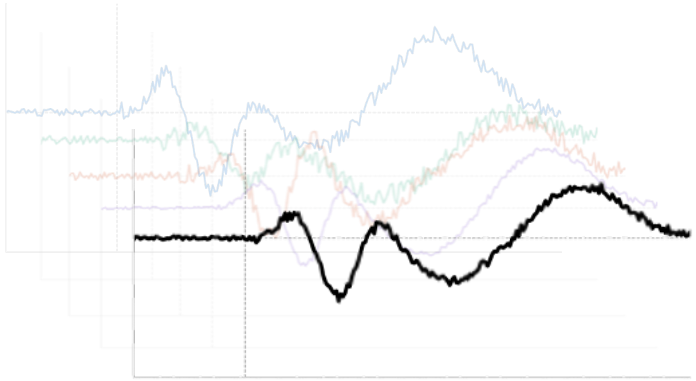


Predictive modelling framework

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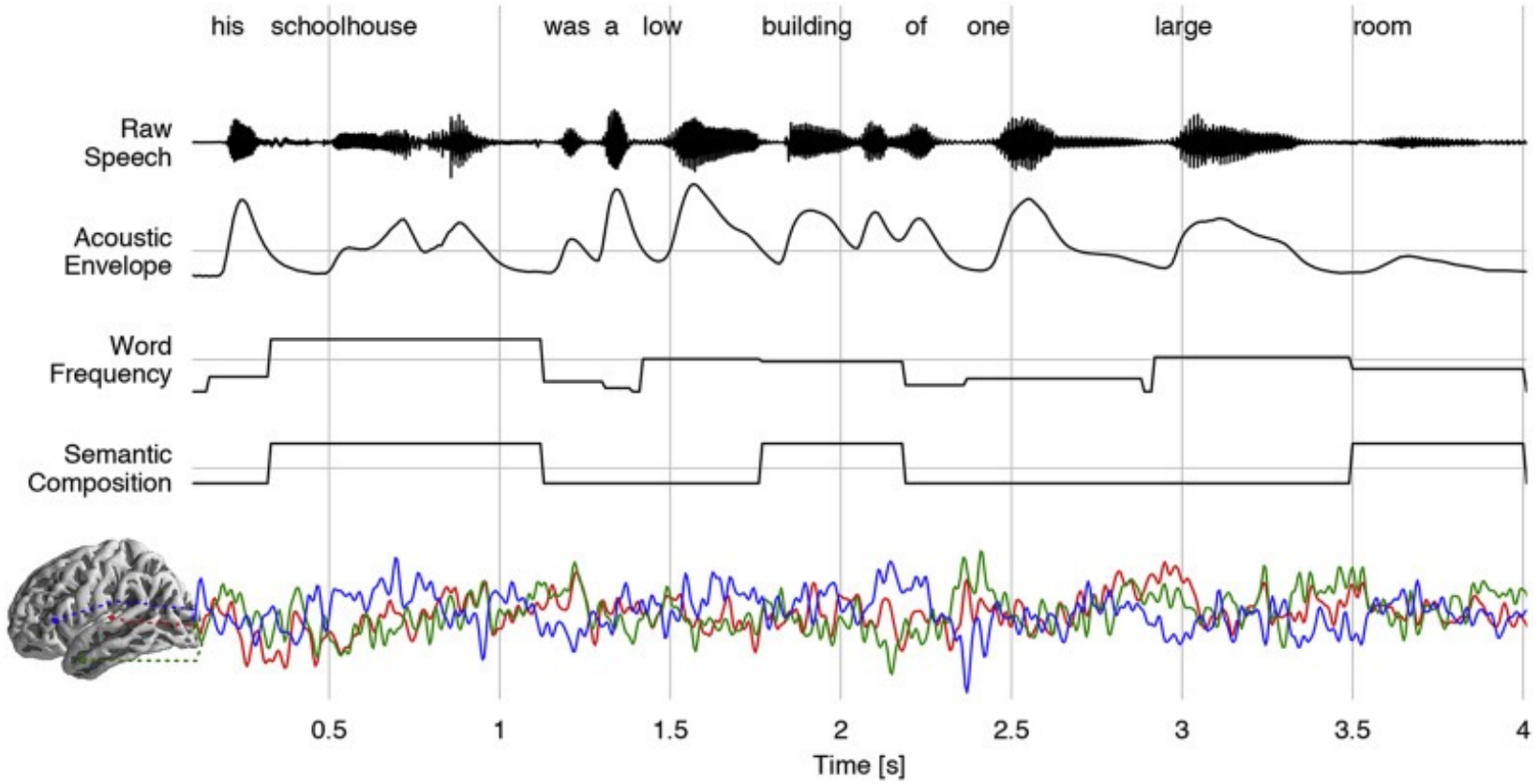
Predictive modelling framework

Beyond the subtraction method

- forces to a restricted set of questions, requiring different stimuli or experiments to study different stimulus features
- forces to artificial task designs

Holdgraf et al. (2017), *Frontiers in Systems Neuroscience*

Predictive modelling framework



Brodbeck et al. (2018), *NeuroImage*

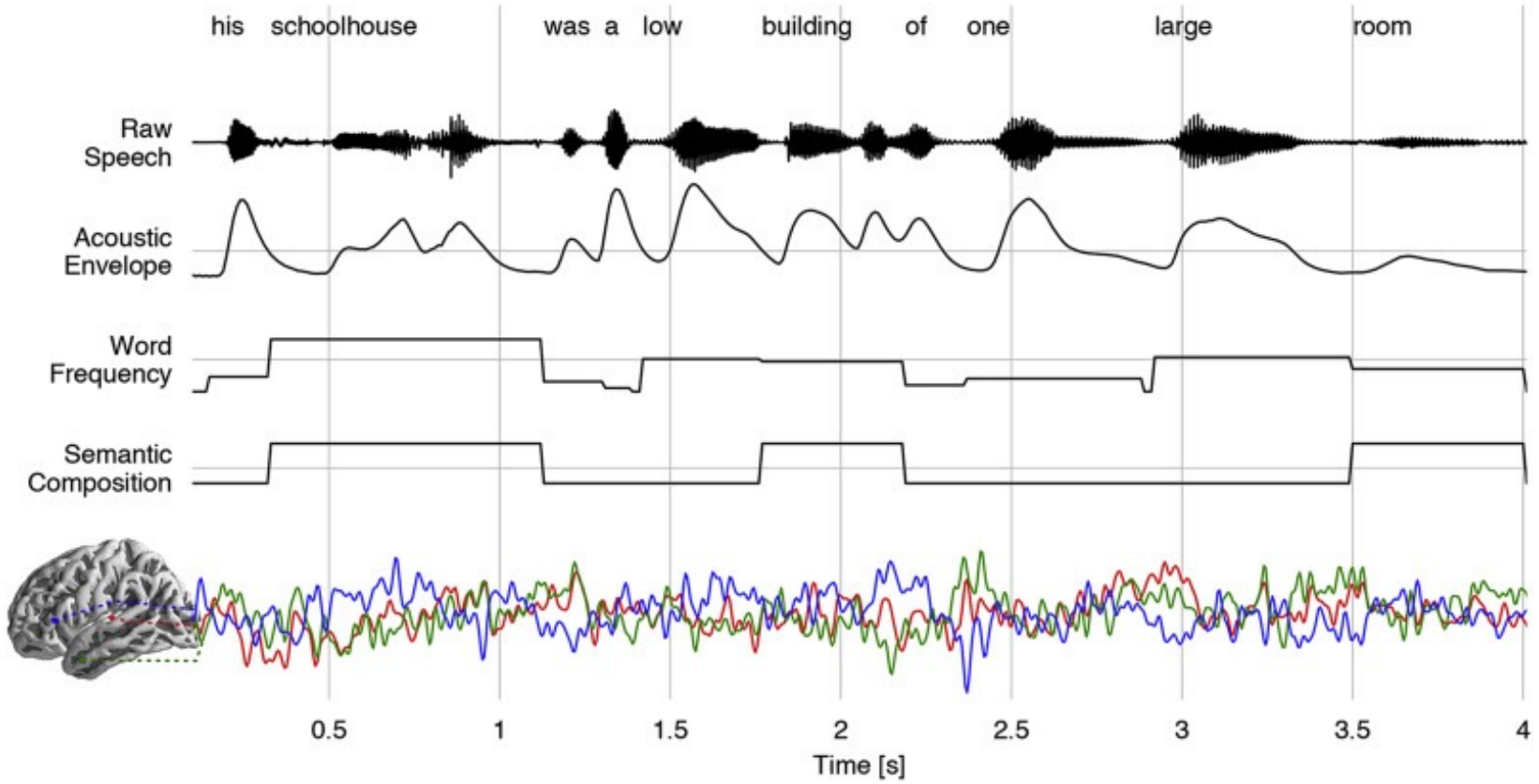
Predictive modelling framework

Advantages:

- more naturalistic stimuli with complex structure
- test multiple hypothesis on the same dataset
- ability to make predictions about new datasets

Holdgraf et al. (2017), *Frontiers in Systems Neuroscience*

Predictive modelling framework



Brodbeck et al. (2018), *NeuroImage*

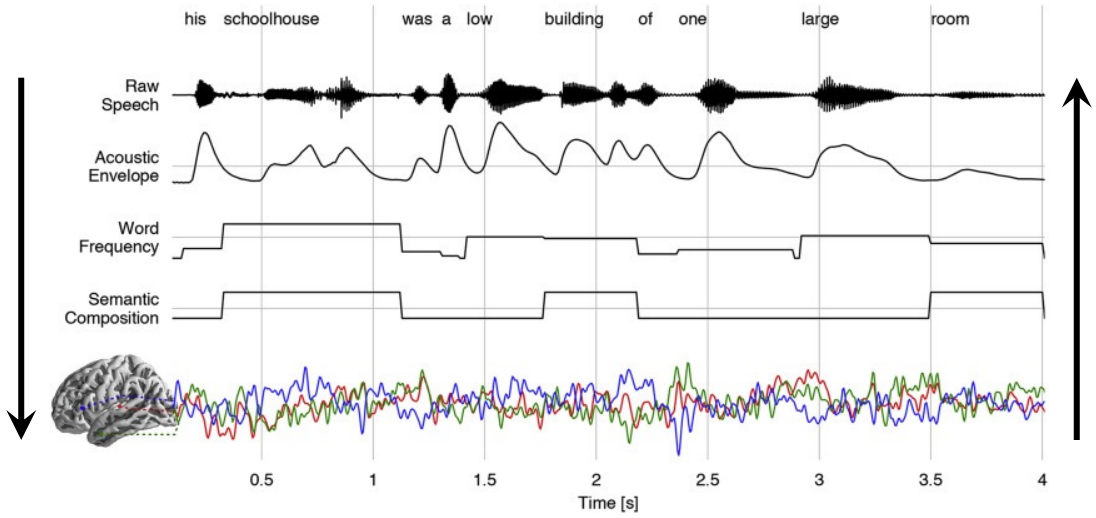
Predictive modelling framework

Encoding (forward modelling)

predicts neural activity from the stimulus features

Decoding (backward modelling)

reconstructs stimulus features from neural activity



Brodbeck et al. (2018), *NeuroImage*

Predictive modelling framework

How does predictive modelling work?

1

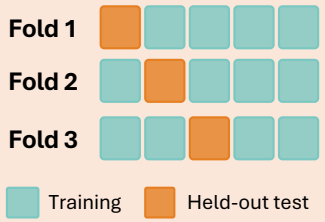
INPUT DATA

$X_{(trials \times channels \times time)}$
$y_{(stimulus\ features)}$

2

SPLIT TRIALS

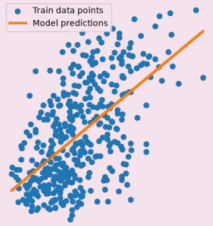
Cross-validation to allow generalization



3

TRAIN MODEL

Scale the predictor and fit ridge regression



4

PREDICT VALUES

Apply the trained model to predict outcomes for unseen trials



5

EVALUATE

Compare predicted and true values

Predictive modelling framework

How does **time-resolved** predictive modelling work?

1

INPUT DATA

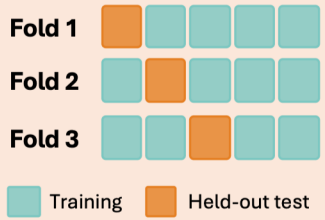
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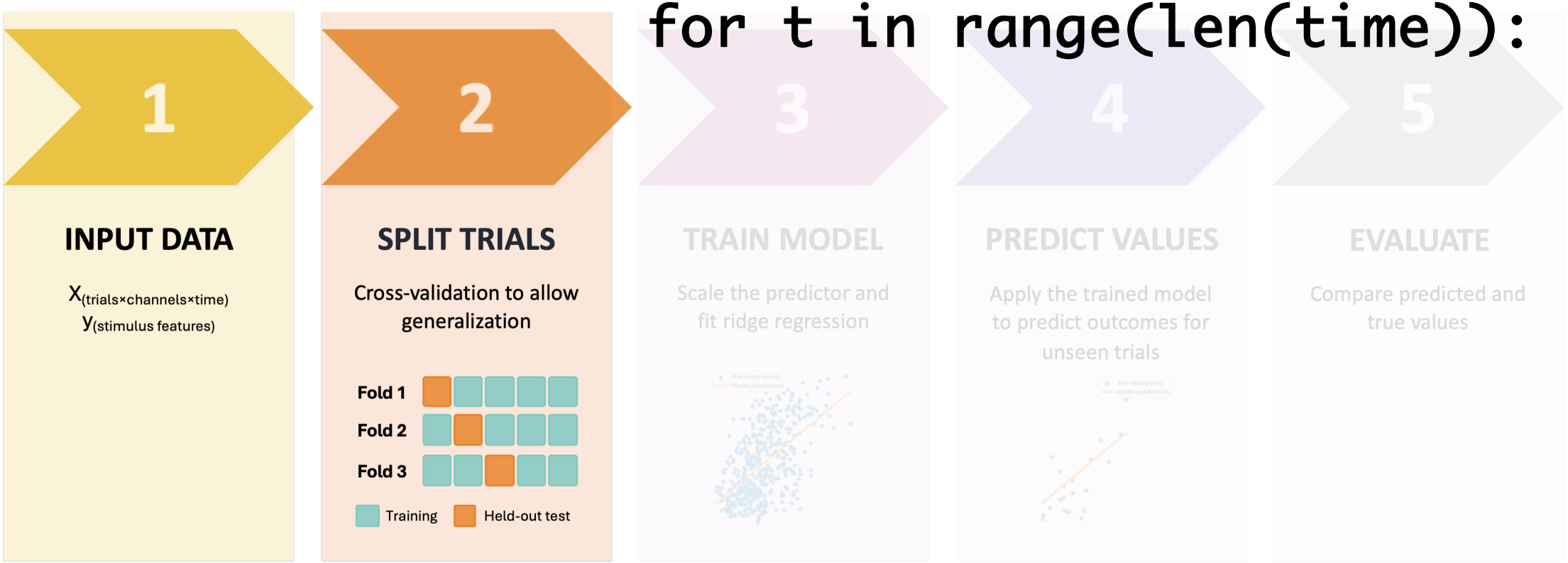
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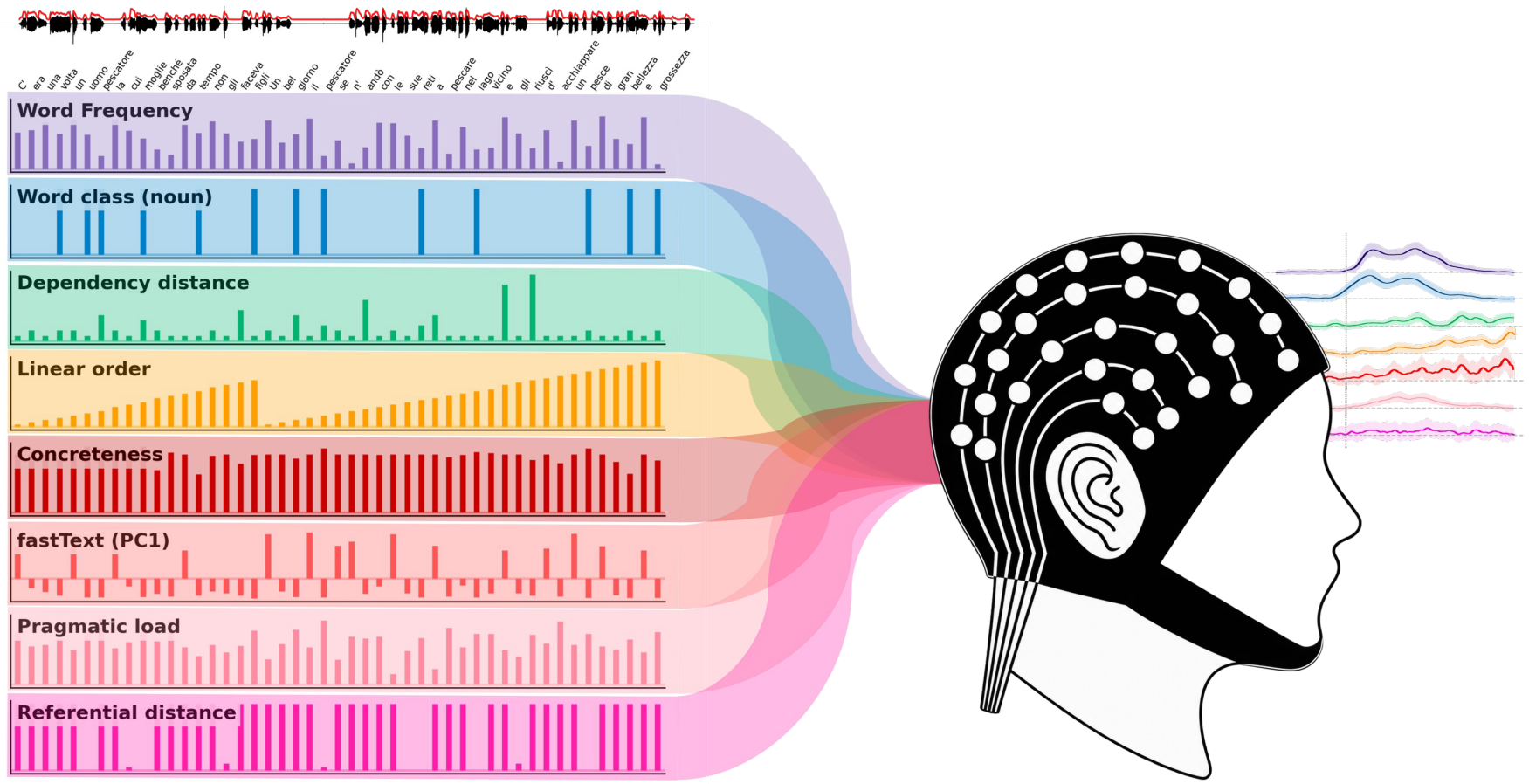
```
for t in range(len(time)):
```



Introduction to the dataset

- **34 participants listened to the Italian folktale** *Il drago dalle sette teste* (Calvino, 1956)
- **EEG activity was recorded** using a 64-channel system
- The narrative transcript was **annotated across more than 50 linguistic dimensions**, spanning acoustic, morpho-syntactic, semantic, and pragmatic features

Introduction to the dataset



Hands-on session

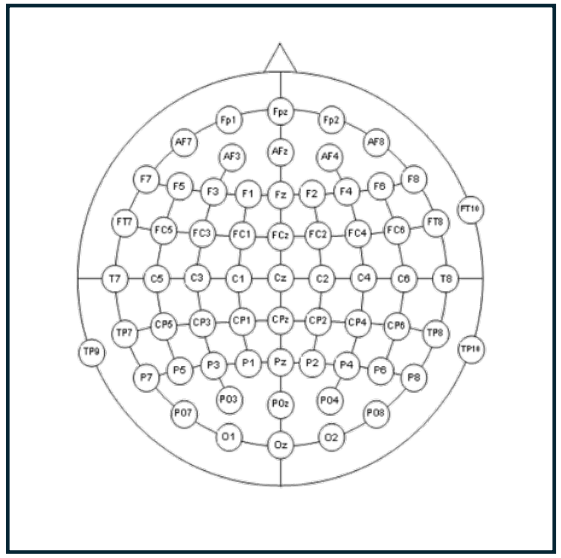
Scan this QR code (or visit this link) to get access to code and data:



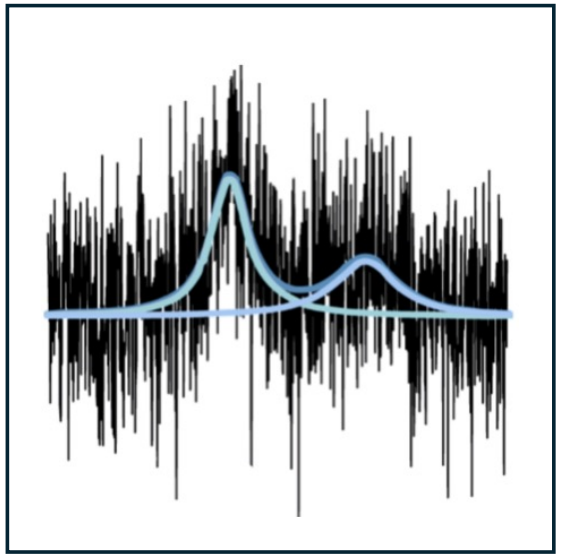
Are multivariate approaches *the* solution?

Multivariate predictive modelling addresses two main problems that cognitive neuroscience dataset pose:

high-dimensional observations



low signal-to-noise ratio



King et al. (2020), *NeuroImage*

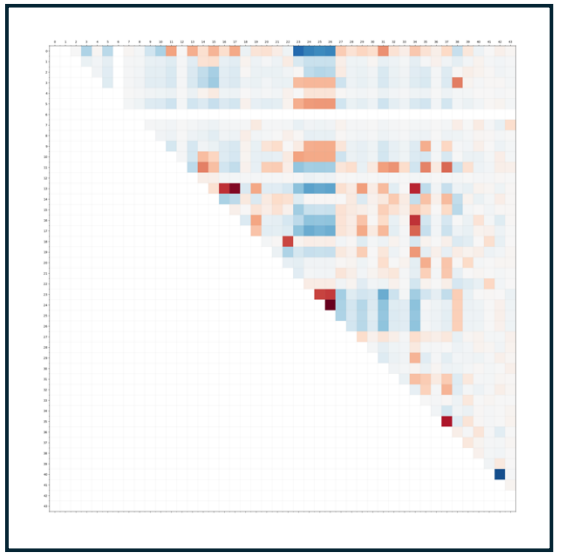
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multicollinear predictors



King et al. (2020), *NeuroImage*

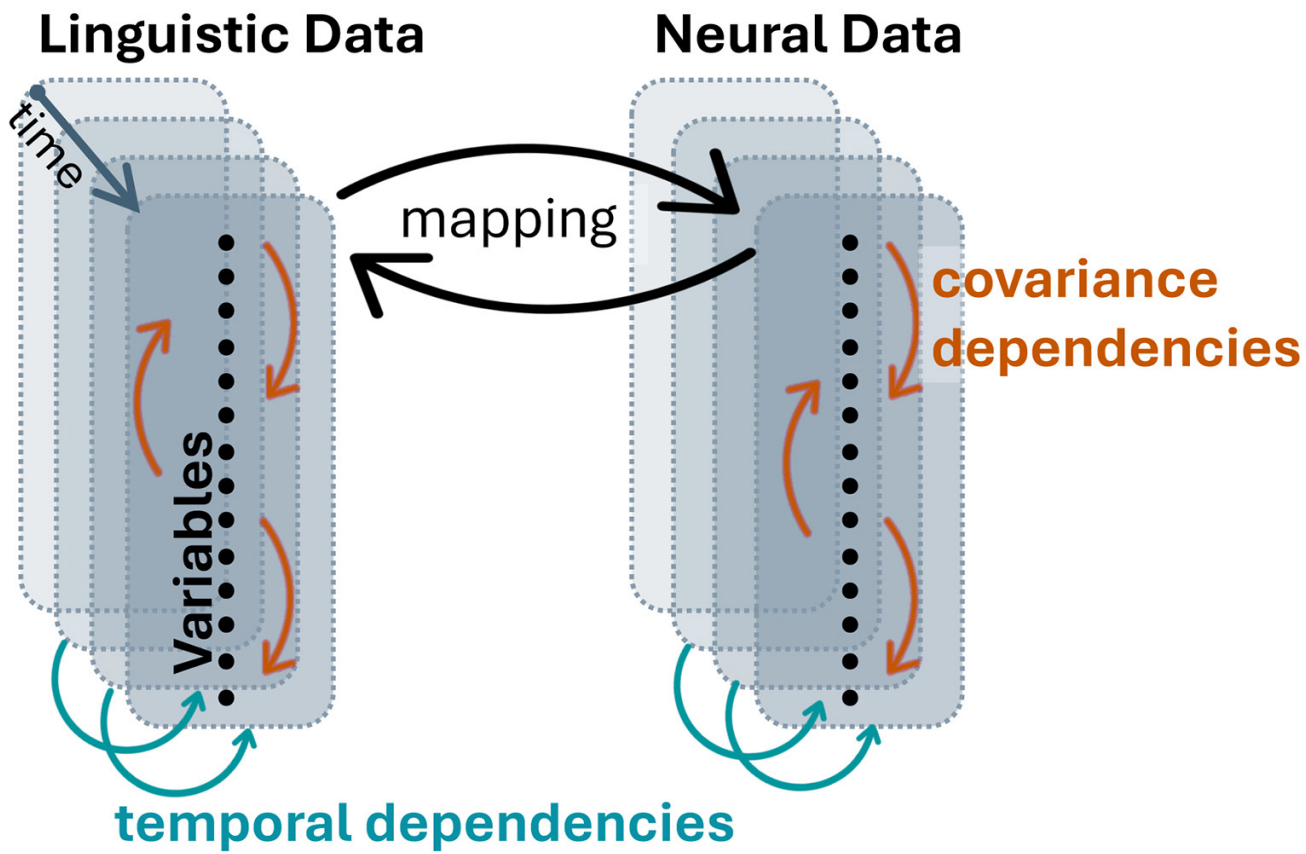
Are multivariate approaches *the* solution?

In cognitive neuroscience of language, in particular, data are inherently characterised with **two statistical dependencies**:

1. covariance
2. temporal dependence

Bouton et al. (2026), *Imaging neuroscience*

Are multivariate approaches *the* solution?



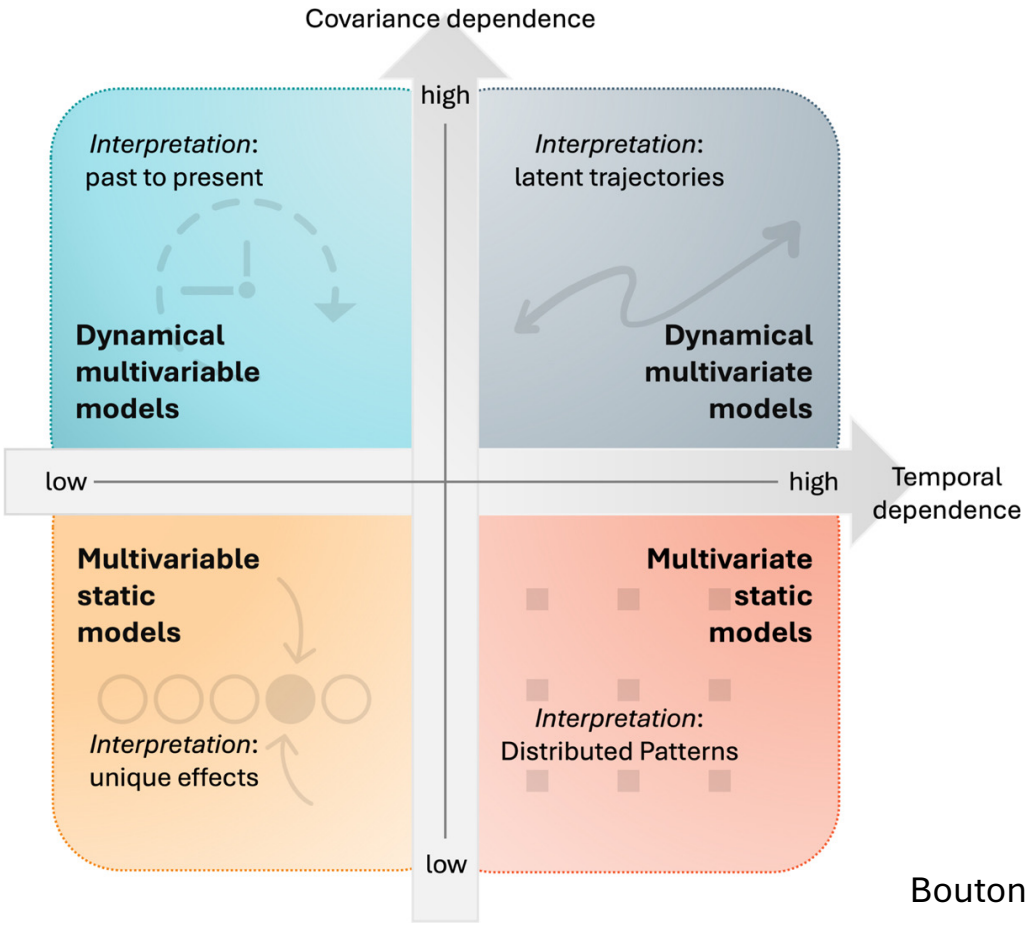
Bouton et al. (2026), *Imaging neuroscience*

Are multivariate approaches *the* solution?

Is multicollinearity between predictors always a problem?

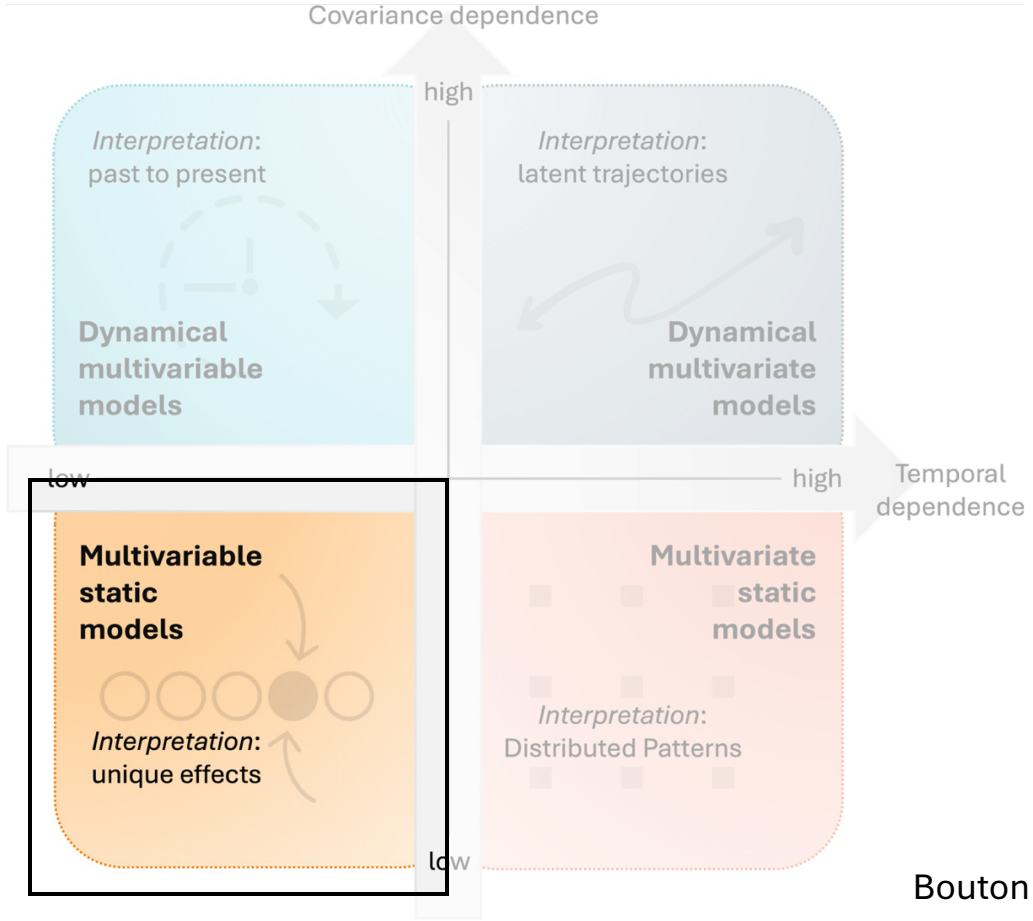
- **It's fine when you want your model to predict data**, for instance in in brain-computer interfaces
- It is more of a problem for driving inferences that require interpretability

Are multivariate approaches *the* solution?



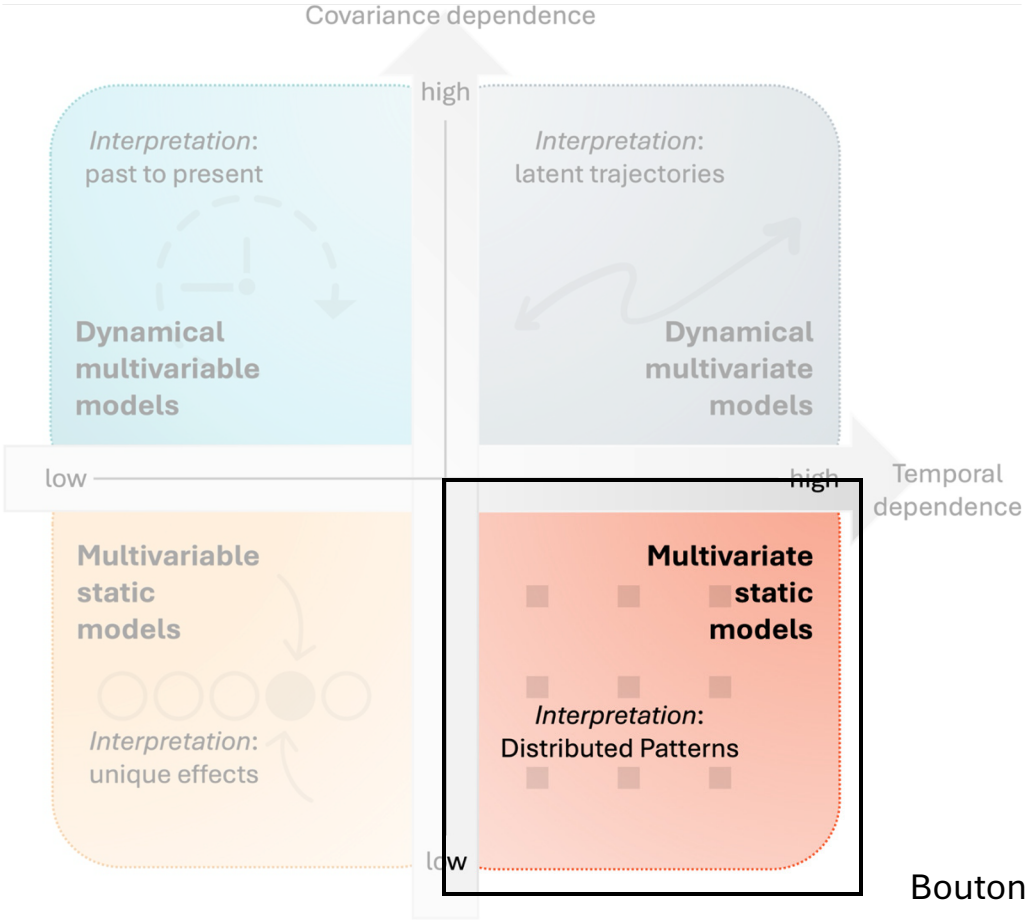
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Are multivariate approaches *the* solution?



Bouton et al. (2026), *Imaging neuroscience*

Are multivariate approaches *the* solution?



Bouton et al. (2026), *Imaging neuroscience*

Back-to-back regression

Goal:

- identify which factors in a set of correlated variables X causally influence high-dimensional observations Y
- combining the benefits of encoding (specificity) and decoding (sensitivity) through a two-step regression approach

King et al. (2020), *NeuroImage*

Back-to-back regression

Problem setup:

- consider Y a multivariate neurophysiological signal
- consider X a set of alleged causes of Y
- consider F the unknown linear relationship between X and Y
- consider S the set of causal influences

$$Y = (XS + \epsilon) \times F$$

King et al. (2020), *NeuroImage*

Back-to-back regression

Algorithm:

1. backward modelling
   i. regress X from Y
   $\hat{G} = Y \sim X$
   ii. generate predicted values $\hat{X}$
   $\hat{X} = \hat{G}Y$
2. forward modelling
   i. regress $\hat{X}$ from X
   $\hat{H} = X \sim \hat{X}$
   ii. take the diagonal of $\hat{H}$
   $\hat{S} = diag(\hat{H})$

$$\hat{W} = X\hat{S} \sim Y$$

King et al. (2020), *NeuroImage*

Back-to-back regression

Algorithm:

1. backward modelling
   i. regress X_1 from Y_1

      $\hat{G} = Y_1 \sim X_1$

   ii. generate predicted values $\hat{X}$

      $\hat{X} = \hat{G} Y_1$

2. forward modelling
   i. regress $\hat{X}$ from X_2

      $\hat{H} = X_2 \sim \hat{G} Y_2$

   ii. take the diagonal of $\hat{H}$

      $\hat{S} = diag(\hat{H})$

$$\hat{W} = X\hat{S} \sim Y$$

King et al. (2020), *NeuroImage*

Back-to-back regression

Feature contribution evaluation:

- a normal Pearson r correlation cannot be computed with non-simulated data, because S is not known
- instead, the evaluation is performed by calculating the difference in the prediction given all features X versus all-but-one feature $X_{_i}$

$$\Delta r_i = r_{full} - r_{knockout(i)}$$

King et al. (2020), *NeuroImage*

Glossary

- **Machine learning**: learning patterns from data to make predictions or decisions without being explicitly programmed
- **Classification**: identifying which category an object belongs to
- **Regression**: predicting a continuous-valued attribute associated with an object
- **Cross-validation**: evaluating a model's performance by training and testing it on different subsets of data
- **Scaling**: transforming numerical features to a common scale
- **Estimator**: an object that learns from data using the fit() method to build a model

scikit-learn – Machine Learning in Python



Thank you :)

and also thanks to the PONS team!

If you wanna stay connected

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